RESEARCH PROGRAMME THROUGH ALPHA: EXACT PAIRING, ENERGY AND ANALYTIC OBLIGATIONS

Independent chapter of Version 12. The results below do not constitute a proof of RH.

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1. Research programme through alpha: exact pairing and analytic obligations

This independently readable chapter states the research programme, proves its intermediate identities, and compares its remaining obligations with established approaches. The comparison concerns the cited works, not every proposed RH proof. An equivalence theorem, a conditional theorem, and a speculative model are not all failed mathematics.

The active route is exact cancellation between sign-specific ordinal pairs, retaining unfinished pairs at every integer cutoff. No estimate of RH strength is assumed. The prime number theorem (PNT) and elementary squarefree counting are used explicitly as unconditional background. Statements are separated into proved results, classical transfers, unresolved targets, and excluded universal mechanisms.

1.1. Step 0: independent notation and target.

Let μ be the usual Möbius function, p_j the jth prime, and $p_0 = 0$. For real $x \geq 1$ define the step functions

$$N_{\pm}(x) = \#\{n \leq x : \mu(n) = \pm 1\}, \quad Q(x) = N_+(x) + N_-(x), \quad M(x) = N_+(x) - N_-(x). \tag{1.1}$$

The transform and its cumulative sum are

$$S(n) = \begin{cases} p_{N_+(n)}, & \mu(n) = 1, \\ -p_{N_-(n)}, & \mu(n) = -1, \\ 1, & \mu(n) = 0, \end{cases} \qquad SS(x) = \sum_{n \leq x} S(n). \tag{1.2}$$

Set $w_j = p_j - j$ and $D_w(m) = \sum_{j=1}^{m} w_j$, with $D_w(0) = 0$. For $\mu(n) \neq 0$, let $j(n)$ be the occurrence number within its own sign. The index $j(n)$ is not used at nonsquarefree integers. Define

$$\alpha(s) = \sum_{n \geq 1} S(n) n^{-s}, \qquad \delta(s) = \sum_{\substack{n \geq 1 \\ \mu(n) \neq 0}} \mu(n) w_{j(n)} n^{-s}. \tag{1.3}$$

Let $s_1 < s_2 < \cdots$ enumerate the squarefree integers. Write

$$\begin{aligned} &\varepsilon_q = \mu(s_q), && R(q) = \sum_{\ell \leq q} \varepsilon_\ell = M(s_q), \qquad R(0) = 0, \\ &\alpha_j = \text{ordinal position of the } j\text{th } +1, && \beta_j = \text{ordinal position of the } j\text{th } -1, \\ &a_j = s_{\alpha_j}, && b_j = s_{\beta_j}, \\ &d_j = \beta_j - \alpha_j, && \Delta_j = b_j - a_j. \end{aligned} \tag{1.4}$$

The function $\alpha(s)$ differs from the ordinal α_j. The physical positions a_j, b_j avoid a clash with the analytic blocks $A(s), B(s)$. Never identify d_j, Δ_j, and w_j. The notation D_w also differs from a lifetime generating function $D(z)$. Finally set

$$c_0 = \frac{3}{\pi^2}, \qquad \Theta = \sup_{\rho \text{ nontrivial}} \Re\rho, \qquad \mathcal{H} = \{s : \Re s > 3/2\}. \tag{1.5}$$

The target is exclusion of nontrivial zeros with $\Re\rho > 1/2$. The functional equation supplies the other half of the strip.

1.2. Step 1: decomposition and genuine pole statements.

Proposition 1.1 (Exact decomposition and domains)**.** *The series in* (1.3) *converge absolutely for* $\Re s > 2$*, and in that domain*

$$\boxed{\alpha(s) = A(s) + B(s) + C(s) + \delta(s),} \tag{1.6}$$

where

$$A(s) = \zeta(s) - \frac{\zeta(s)}{\zeta(2s)}, \tag{1.7}$$

$$B(s) = c_0 \frac{s}{(s-1)\zeta(s-1)}, \tag{1.8}$$

$$C(s) = \frac{s}{2}\int_1^\infty M(x)(Q(x)+1-2c_0x)x^{-s-1}\,dx. \tag{1.9}$$

The blocks A, B are meromorphic on $\mathbb{C}$. The integral for C is holomorphic on $\mathcal{H}$ unconditionally; more precisely, its domain includes $\Re s > \Theta + 1/2$.

Proof. The bound $p_j \ll j\log(2j)$ majorises both coefficient sequences by $O(n\log(2n))$. The signed ordinal coefficients have the partial sum

$$\sum_{\substack{n\le x\\ \mu(n)\neq 0}} \mu(n)j(n) = \frac{N_+(N_++1) - N_-(N_-+1)}{2} = \frac{M(x)(Q(x)+1)}{2}. \tag{1.10}$$

Partial summation gives the integral of this expression against $sx^{-s-1}\,dx$. Splitting $Q+1 = 2c_0x + (Q+1-2c_0x)$ and using

$$\int_1^\infty M(x)x^{-z-1}\,dx = \frac{1}{z\zeta(z)} \qquad (\Re z > 1) \tag{1.11}$$

gives $B + C$. The zero row contributes A. The unconditional estimates $Q(x) = 2c_0x + O(\sqrt{x})$ and $|M(x)| \le x$ give local uniform absolute convergence of the C integral for $\Re s > 3/2$. The classical bound $M(x) = O_\epsilon(x^{\Theta+\epsilon})$ gives the refined domain [6]. □

Proposition 1.2 (Poles of the ordinal main block)**.** *A nontrivial zero ρ of ζ gives a pole of B at $1+\rho$ with the same order. For a simple zero,*

$$\operatorname*{Res}_{s=1+\rho} B(s) = c_0 \frac{1+\rho}{\rho\zeta'(\rho)}. \tag{1.12}$$

There is also a simple pole at $s = 1$ of residue $-2c_0$, and the shifted trivial-zero poles $s = 1-2k$, $k \ge 1$. The point $s = 2$ is a zero, not a pole.

Proof. Use the local factorisation of $\zeta(s-1)$ at its zeros. The prefactor $c_0 s/(s-1)$ is finite and nonzero at every shifted zero. At $s = 1$ use $\zeta(0) = -1/2$; at $s = 2$ use the pole of ζ at 1. □

No corresponding pole count for the full α has been proved. Critical-line zero theorems transfer to B, not automatically to α: continuation and noncancellation of the other blocks must first be checked. Holomorphy of δ alone does not remove a pole of B from α. Nor does extending δ across $3/2$ extend C across that line.

1.3. **Step 2: exact pairing without linearisation.**

Proposition 1.3 (Pairs, oriented integrals, and unit cuts)**.** *For $\Re s > 2$,*

$$\delta(s) = \sum_{j\ge 1} w_j(a_j^{-s} - b_j^{-s}). \tag{1.13}$$

For each pair and every complex s,

$$a_j^{-s} - b_j^{-s} = s\int_{a_j}^{b_j} x^{-s-1}\,dx. \tag{1.14}$$

With

$$u_q = \frac{q+R(q)}{2}, \quad v_q = \frac{q-R(q)}{2}, \quad U(q) = D_w(u_q) - D_w(v_q), \tag{1.15}$$

the initial half-plane has the representation

$$\boxed{\delta(s) = \sum_{q\geq 1} U(q)(s_q^{-s} - s_{q+1}^{-s}).} \tag{1.16}$$

Proof. Absolute convergence allows occurrence matching. Differentiating x^{-s} proves the oriented integral, including $s = 0$. Telescoping gives

$$a_j^{-s} - b_j^{-s} = \operatorname{sgn}(d_j) \sum_{q=\min(\alpha_j,\beta_j)}^{\max(\alpha_j,\beta_j)-1} (s_q^{-s} - s_{q+1}^{-s}). \tag{1.17}$$

At cut q the open labels are exactly $\min(u_q, v_q) < j \leq \max(u_q, v_q)$, all with orientation $\operatorname{sgn} R(q)$. Their signed weight is $U(q)$. The bound $|U(Q(x))| \ll x^2 \log(2x)$ gives an integrable absolute majorant $|s|x^{1-\Re s}\log(2x)$ when $\Re s > 2$, justifying the rearrangement. □

In particular,

$$U(q) = \operatorname{sgn} R(q) \sum_{j=\min(u_q,v_q)+1}^{\max(u_q,v_q)} w_j. \tag{1.18}$$

The kernel in (1.16) is local, but its coefficient still contains every long open pair. At one cut all these contributions have the same sign. Useful cancellation must occur across different cuts.

PNT already gives $\alpha_j, \beta_j \sim 2j$ and $a_j, b_j \sim j/c_0$. Thus $d_j = o(j)$ and $\Delta_j = o(j)$ unconditionally: eventually excluding a macroscopic relative separation does not require RH. This relative smallness supplies no square-root cancellation, and a Taylor argument would still need uniform summable errors. The exact formulas avoid that requirement entirely.

1.4. **Step 3: the nonpolynomial duration observable.** For this step take a balanced finite word of length $Q = 2J$, so $R(Q) = 0$. All its J pairs are complete. Define half-open cut intervals and duration tails:

$$I_j = [\min(\alpha_j, \beta_j), \max(\alpha_j, \beta_j)), \qquad V_J(r) = \sum_{j=1}^{J} (|d_j| - r)_+. \tag{1.19}$$

Proposition 1.4 (Two-cut identity)**.** *For $0 \leq r < Q$,*

$$\boxed{V_J(r) = \frac{1}{2} \sum_{q=0}^{Q-r-1} (|R(q) + R(q+r)| - r)_+.} \tag{1.20}$$

For $r \geq Q$ use an empty sum. In particular,

$$V_J(0) = \sum_{q=0}^{Q-1} |R(q)| \geq J. \tag{1.21}$$

For $0 < z < 1$,

$$\Phi_J(z) := \sum_{j=1}^{J} z^{|d_j|} = J - (1-z)V_J(0) + (1-z)^2 \sum_{r=1}^{Q-1} z^{r-1} V_J(r). \tag{1.22}$$

Proof. Using $N_\pm(q) = (q \pm R(q))/2$ in the ordinal clock, pairs opening with plus and crossing both cuts have labels $N_-(q+r) < j \leq N_+(q)$. Those opening with minus have labels $N_+(q+r) < j \leq N_-(q)$. Their disjoint counts give

$$C_r(q) = (N_+(q) - N_-(q+r))_+ + (N_-(q) - N_+(q+r))_+ = \tfrac{1}{2}(|R(q) + R(q+r)| - r)_+. \tag{1.23}$$

An interval of length ℓ contains both cuts for $(\ell - r)_+$ choices of q. Double counting proves (1.20). The case $r = 0$ and $|d_j| \geq 1$ give (1.21). Finally, sum the finite identity

$$z^\ell = 1 - (1-z)\ell + (1-z)^2 \sum_{r=1}^{\ell-1} z^{r-1}(\ell - r) \tag{1.24}$$

over the pairs. □

Proposition 1.5 (Order survives a one-time height histogram)**.** *The words* $\mathrm{I} = ++-+--+-$ *and* $\mathrm{II} = ++--++--$ *have the same height histogram but different duration tails and transforms.*

Proof. Their height lists are $(1,2,1,2,1,0,1,0)$ and $(1,2,1,0,1,2,1,0)$: both contain two zeros, four ones, and two twos. Their durations are $(2,3,2,1)$ and $(2,2,2,2)$, respectively. Therefore

$$V_{\mathrm{I}}(2) = 1, \quad V_{\mathrm{II}}(2) = 0, \qquad \Phi_{\mathrm{I}}(z) - \Phi_{\mathrm{II}}(z) = z(1-z)^2 > 0. \tag{1.25}$$

Their quadratic duration sums also differ, 18 versus 16. Not every quadratic functional is insensitive to order. □

Order sensitivity is not yet arithmetic specificity. If F_+, F_- are the distribution functions of the empirical probability measures $J^{-1}\sum_j \delta_{\alpha_j}$ and $J^{-1}\sum_j \delta_{\beta_j}$, then

$$\frac{V_J(0)}{J} = \frac{1}{J}\sum_j |\alpha_j - \beta_j| = \int_{\mathbb{R}} |F_+(t) - F_-(t)|\, dt. \tag{1.26}$$

This is the classical one-dimensional monotone Wasserstein transport identity [14]. Neither monotone matching nor its first absolute cost is claimed as new.

1.4.1. *What fixed-lag arithmetic input actually controls.* The squarefree-clock two-point theorem obtained from Tao's logarithmic two-point theorem [5] gives, for every fixed $r \geq 1$,

$$\frac{1}{\log X}\sum_{q \leq X} \frac{(R(q+r) - R(q))^2}{q} \longrightarrow r. \tag{1.27}$$

The clock transfer keeps a physical separation and a squarefree-sieve modulus fixed when applying the two-affine-form theorem. The error from omitted prime squares tends to zero as the sieve cutoff subsequently tends to infinity. Large physical separations have vanishing logarithmic tail because $\sum_{q \leq X}(s_{q+r} - s_q) = O_r(X)$ by telescoping. Expanding the increment square then gives r diagonal terms and finitely many correlations. These successive limits do not give uniformity in growing r.

The precise relation to the duration observable is

$$\left| C_r(q) - \left(|R(q)| - \frac{r}{2} \right)_+ \right| \leq \frac{1}{2}|R(q+r) - R(q)|. \tag{1.28}$$

It follows from the 1-Lipschitz property of absolute value and positive part. Cauchy–Schwarz gives

$$\limsup_{X \to \infty} \frac{1}{\log X}\sum_{q \leq X} \frac{|C_r(q) - (|R(q)| - r/2)_+|}{q} \leq \frac{\sqrt{r}}{2}. \tag{1.29}$$

The unestimated main term depends on the absolute height. A theorem on fixed increments is therefore not automatically a bound for $V_J(r)$, even for fixed r. Moreover V_J is unsigned and lacks the weights w_j; both transfers to δ require additional argument.

Remark 1.6 (Height, area, and prime energy are different). Let $H = \max_{q \leq Q} |R(q)|$. The H pairs active at a cut of height $\pm H$ each have duration at least H: for the tth active label, at least $H - t$ steps separate its opening from the cut and at least t steps separate the cut from its closing. Hence

$$V_J(r) \geq H(H - r)_+. \tag{1.30}$$

In particular $V_J(0)$ is an area, not a height. The bound $V_J(0) \ll_\epsilon J^{1/2+\epsilon}$ for every $\epsilon > 0$ is impossible because $V_J(0) \geq J$. A square-root-plus-ϵ height bound instead gives an area bound $O_\epsilon(J^{3/2+\epsilon})$. The cubic negative-excursion prime-separation energy is a different functional.

1.5. **Step 4: holomorphy for a fixed duration cutoff.**

Theorem 1.7 (Unconditional bounded-duration holomorphy)**.** *For every fixed $L \geq 1$ the series*

$$\delta_{\leq L}(s) = \sum_{|d_j| \leq L} w_j (a_j^{-s} - b_j^{-s}) \tag{1.31}$$

converges normally by pairs in $\Re s > 1$ and is holomorphic there. For $\sigma = \Re s > 1$,

$$\sum_{|d_j| \leq L} w_j |a_j^{-s} - b_j^{-s}| \ll L|s| \left(\frac{\log 2}{\sigma - 1} + \frac{1}{(\sigma-1)^2} \right). \tag{1.32}$$

The proof uses no Mertens estimate, density theorem, or linearisation.

Proof. At physical cut x, put $q = Q(x)$. An active ordinal interval of length at most L opened within the last L positions. There are at most L such intervals, and each has label $j \leq q \leq x$, so $w_j \ll x \log(2x)$. The exact integral and Tonelli give

$$\begin{aligned} \sum_{|d_j| \leq L} w_j |a_j^{-s} - b_j^{-s}| &\leq |s| \int_1^\infty x^{-\sigma-1} \sum_{\substack{|d_j| \leq L \\ \min(a_j,b_j) \leq x < \max(a_j,b_j)}} w_j \, dx \\ &\ll L|s| \int_1^\infty x^{-\sigma} \log(2x) \, dx. \end{aligned} \tag{1.33}$$

The last integral equals the parenthesis in (1.32). On any compact set use its minimum real part and maximum $|s|$ to obtain a summable majorant for the suprema of the paired terms. □

This theorem also holds for arbitrary sign-occurrence lists in an increasing integer clock and weights $|w_j| \ll j \log(2j)$. Its generality is useful, but it is not an arithmetic cancellation theorem.

For $\Re s > 2$, dominated convergence gives $\delta_{\leq L} \to \delta$ locally uniformly as $L \to \infty$. The displayed bound grows with L; it does not justify this limit on $\mathcal{H}$.

1.6. **Step 5: long pairs and the sign involution.** Large rank and large duration are different: an alternating word has $|d_j| = 1$ at every rank, however large w_j becomes.

Proposition 1.8 (Universal antisymmetry does not delete the long tail)**.** *Sign reversal and ordinal matching alone do not justify discarding all pairs of duration greater than a fixed L.*

Proof. For $H > L$, take the auxiliary balanced word $(+1)^H(-1)^H$ on any increasing positive clock. Every duration equals H, but for real $\sigma > 0$,

$$\delta_{\leq L}(\sigma) = 0, \qquad \sum_{j=1}^{H} w_j (s_j^{-\sigma} - s_{H+j}^{-\sigma}) > 0. \tag{1.34}$$

There are no short pairs. This disproves the claimed universal deletion, not a possible cancellation specific to the genuine Möbius word. □

The global sign flip gives

$$R \mapsto -R, \qquad d_j \mapsto -d_j, \qquad \delta(s) \mapsto -\delta(s). \tag{1.35}$$

It has no action on s and gives no equation under $s \mapsto 3 - s$. The zero row of α is unchanged, so the whole transform does not simply change sign either.

If a universal functional satisfies $F(\varepsilon) \geq c\,\Psi(\|d(\varepsilon)\|_\infty)$ for both signs, then its evenisation $[F(\varepsilon) + F(-\varepsilon)]/2$ has the same lower bound. Thus odd cancellation alone cannot explain such universal coercivity. This does not exclude all even, nonpolynomial, or arithmetic-specific functionals.

1.7. **Step 6: the precise original-order target.** For integer X let

$$\delta_X(s) = \sum_{j\geq 1} w_j(\mathbf{1}_{a_j\leq X}a_j^{-s} - \mathbf{1}_{b_j\leq X}b_j^{-s}). \tag{1.36}$$

This is a finite sum in the original integer ordering.

1.7.1. *The unfinished-pair term.* Put $m_X = \min(N_+(X), N_-(X))$ and $h_X = |N_+(X) - N_-(X)|$. Exactly,

$$\delta_X(s) = \sum_{j\leq m_X} w_j(a_j^{-s} - b_j^{-s}) + E_X(s), \tag{1.37}$$

where

$$E_X(s) = \begin{cases} \displaystyle\sum_{j=m_X+1}^{m_X+h_X} w_j a_j^{-s}, & N_+(X) > N_-(X), \\ \displaystyle -\sum_{j=m_X+1}^{m_X+h_X} w_j b_j^{-s}, & N_-(X) > N_+(X), \\ 0, & N_+(X) = N_-(X). \end{cases} \tag{1.38}$$

At a real argument all terms of E_X have the same sign. Convergence along balanced prefixes or after completing every pair does not justify convergence at arbitrary integer cutoffs.

1.7.2. *Unresolved Cauchy estimate.* The central target is, for every compact $K \subset \mathcal{H}$,

$$\boxed{\lim_{X\to\infty} \sup_{\substack{Y\in\mathbb{N}\\ Y\geq X}} \sup_{s\in K} |\delta_Y(s) - \delta_X(s)| = 0.} \tag{1.39}$$

A concrete sufficient estimate is $|\delta_Y(s) - \delta_X(s)| \leq \eta_K(X)$ for all such Y, s, where an independently controlled $\eta_K(X)$ tends to zero. Complete short pairs, complete long pairs, and pairs cut by either endpoint must be handled with compatible uniformity.

This is locally uniform convergence of original partial sums, not absolute convergence of individual integer terms, whose threshold remains 2. A different grouping that gives analytic continuation is not automatically a proof of (1.39).

1.8. **Step 7: the rigorous transfer to RH.**

Proposition 1.9 (The exact coefficient sum and its scale)**.** *Let* $W(x) = \sum_{n\leq x,\ \mu(n)\neq 0} \mu(n) w_{j(n)}$. *Then*

$$W(x) = D_w(N_+(x)) - D_w(N_-(x)) = SS(x) - \lfloor x \rfloor + Q(x) - \frac{M(x)(Q(x)+1)}{2}. \tag{1.40}$$

There is $e(x) \to 0$ *such that, for sufficiently large* x,

$$W(x) = c_0 x \log x\, M(x)(1 + e(x)), \tag{1.41}$$

with both sides zero when $M(x) = 0$.

Proof. The difference of cumulative weights gives the first identity, and (1.10) gives the second. PNT implies $Q(x) \sim 2c_0 x$ and $M(x) = o(x)$, hence both counters are $c_0 x(1 + o(1))$. Uniformly between them, $w_j = p_j - j \sim j \log j \sim c_0 x \log x$. The block has signed length $M(x)$. This uses PNT, not an RH-sized bound. □

Theorem 1.10 (Original-order criterion and RH)**.** *The Cauchy target* (1.39) *is equivalent to RH. It is also equivalent to convergence of the ordinary series* $\delta(\sigma)$ *for every real* $\sigma > 3/2$.

Proof. Real convergence and Kronecker's lemma give $W(x) = o(x^\sigma)$. Consequently (1.41) gives

$$M(x) = o\left(\frac{x^{\sigma-1}}{\log x}\right). \tag{1.42}$$

Taking $\sigma = 3/2 + \epsilon/2$ for each $\epsilon > 0$ yields the Littlewood criterion for RH [3, 6]. Its needed direction follows because these estimates make $z\int_1^\infty M(x)x^{-z-1}\,dx$ holomorphic for $\Re z > 1/2$. It agrees with $1/\zeta(z)$ for $\Re z > 1$, so no zero is possible to the right of $1/2$; symmetry completes RH.

Conversely, under RH, $M(x) = O_\eta(x^{1/2+\eta})$ gives $W(x) = O_\eta(x^{3/2+\eta}\log x)$. For compact $K \subset \mathcal{H}$ choose $0 < \eta < \min_{s\in K}(\Re s - 3/2)$. Partial summation gives exactly

$$\delta_Y(s) - \delta_X(s) = W(Y)Y^{-s} - W(X)X^{-s} + s\int_X^Y W(x)x^{-s-1}\,dx. \tag{1.43}$$

All terms tend to zero uniformly in $Y \geq X$ and $s \in K$. □

This is a classical transfer in new coefficients, not a proof of its unresolved premise. In the forward direction the Mertens estimate is a conclusion. Assuming it while proving the premise would be circular.

The standard abscissa formula and the classical identification of the Mertens growth exponent with Θ also yield

$$\sigma_c(\alpha) = \sigma_c(\delta) = 1 + \Theta. \tag{1.44}$$

Indeed W has growth exponent $1 + \Theta$; the signed prime part of SS has the same exponent, whereas $\lfloor x \rfloor - Q(x) = O(x)$. The abscissae refer to the original ordering, not to arbitrary paired rearrangements. An abscissa is not automatically a line of poles.

The ordinary series $\alpha(3/2)$ diverges. Otherwise Kronecker's lemma would give $SS(x) = o(x^{3/2})$ and the same signed prime-block asymptotic would force $M(x) = o(\sqrt{x}/\log x)$, contrary to $M(x) = \Omega_\pm(\sqrt{x})$ [1, 2]. The stronger constants of the Mertens-conjecture disproof are unnecessary. Divergence of this series does not decide whether a continuation could have a finite value at the real point $3/2$.

1.9. **Step 8: the separate continuation-and-poles route.** The meromorphic route has a different sufficient objective:

$$\boxed{\begin{aligned}&\text{RH} \iff B \in \mathcal{O}(\mathcal{H})\\ &\quad\iff \alpha - \delta \text{ admits holomorphic continuation to } \mathcal{H}.\end{aligned}} \tag{1.45}$$

The first equivalence follows from Proposition 1.2 and zeta symmetry; the second uses $A, C \in \mathcal{O}(\mathcal{H})$ and (1.6), initially on $\Re s > 2$. Separate holomorphy of both α and δ suffices; holomorphy of δ alone does not.

Subtracting δ leaves the zero row and the ordinal part. Without a new independent estimate, this returns to a classical shifted reciprocal-zeta problem. There is no assumed global bijection between the poles of the full α and the zeros of ζ.

1.9.1. *8A: contours.* A contour route must establish continuation, control away from poles, and a count excluding poles. Uniform polynomial growth on a strip containing poles is impossible without removing their neighbourhoods or their principal parts. Furthermore,

$$\frac{1}{2\pi i}\oint_\Gamma \frac{\alpha'}{\alpha}\,ds = N_{\text{zeros}}(\Gamma) - N_{\text{poles}}(\Gamma). \tag{1.46}$$

A zero integral is not a zero pole count. A lower bound on a separating curve does not supply the missing interior count. For RH it suffices to work in the open half-plane $\mathcal{H}$; a strip crossing its boundary is not intrinsically necessary.

1.9.2. *8B: positivity and the correctly shifted Li map.* One first needs a completed function with proved divisor, controlled growth, and an appropriate convergent product. For candidate poles $\varpi = 1 + \rho$, the transferred expression is

$$\lambda_n^{\alpha} = \lim_{T\to\infty} \sum_{\substack{\varpi=1+\rho \\ |\Im\varpi|\le T}} \left[1 - \left(\frac{\varpi - 2}{\varpi - 1}\right)^n\right], \tag{1.47}$$

with multiplicities and the displayed summation convention. The map is not an arbitrary $1-\varpi_0/\varpi$:

$$\left|\frac{\varpi - 2}{\varpi - 1}\right| = 1 \quad\Longleftrightarrow\quad \Re\varpi = \frac{3}{2}. \tag{1.48}$$

Li's theorem gives a criterion, not automatic positivity [7]. An ordinal implementation must derive the relevant form and prove every required inequality. Generic Gram positivity is insufficient. The sign involution supplies neither the divisor nor the completion.

1.9.3. *8C: spectral realisation.* A spectral programme would require:

(i) a specified Hilbert space and densely defined self-adjoint operator;
(ii) a justified determinant, regularised when necessary;
(iii) an exact relation between its divisor and the relevant analytic divisor;
(iv) an affine spectral map taking real spectrum to $\Re s = 3/2$.

Reality of a spectrum matters only after the divisor relation is proved. An arbitrary real-spectrum matrix, or an operator defined using already known zero ordinates, does not provide that relation independently.

1.10. **Step 9: corrected rules and excluded shortcuts.**

R1. Do not assume an RH-equivalent estimate for M, SS, W, d_j or a corresponding abscissa while trying to prove it. An independently proved equivalent assertion is not circular; neither is an unconditional bound.
R2. Distinguish original-order convergence, absolute convergence by pairs, and analytic continuation. Retain unfinished-pair boundary terms.
R3. Use the exact integral (1.14). The presence of a displacement in an identity is not circular; assuming an RH-sized bound for it would be.
R4. Do not transfer poles or a functional equation from a single summand to α. Coefficient size alone does not force a symmetry line.
R5. A power-saving prime-sum remainder requires a specified main term. A finite inverse-logarithm expansion does not supply it. This obstructs that approximation scheme, not every possible analytic continuation.
R6. A prime-index block of length $|M(x)|$ is not automatically a physical prime interval of length $\sqrt{x}$. No argument here identifies a single missing ϵ in a prime-interval bound as the sole obstacle.
R7. Fixed-lag logarithmic cancellation does not estimate the absolute height in (1.28), growing lags, or the full signed weighted tail without further proof.
R8. An identity valid for every sign word is not by itself arithmetic specificity. Auxiliary words test a universal mechanism; they are not asserted to be the Möbius sequence.
R9. Quadratic comparison theorems prove their stated comparisons only. Proposition 1.5 refutes the claim that every quadratic functional loses order.
R10. The off-boundary assertion

$$\int_1^{\infty} M(x)^2 x^{-2\sigma-1}\,dx < \infty \quad \text{for every } \sigma > 1/2 \tag{1.49}$$

is equivalent to RH without simplicity of zeros. RH implies finiteness by choosing a smaller exponent. Conversely, Cauchy–Schwarz makes the Mellin integral of M analytic to the right

of 1/2. The stronger weak Mertens condition $\int_1^X M(x)^2 x^{-2}\,dx = O(\log X)$ has additional simplicity and negative-moment consequences [1]. The active route here does not assume a quadratic estimate.

R11. Balanced finite words justify combinatorial identities, not replacement of all integer cutoffs by balanced cutoffs in an infinite analytic argument.

1.11. **Step 10: comparison with established research programmes.** Each comparison separates what the cited work proves from what would still be needed for RH. The missing implication identified here is a logical assessment, not a claim that the programme is impossible or that its authors' valid theorem is erroneous.

1.11.1. *10A: Mertens criteria and conditional moments.* The Mellin identity (1.11) and the Littlewood criterion connect cancellation of μ to zero exclusion [3, 6]. Replacing a Mertens bound by one for SS or W does not independently prove it. The additional occurrence-pair representation is the distinction here, not a weaker convergence threshold.

Ng's distributional and moment theorems explicitly use RH and additional negative-moment hypotheses [1]. Using their conclusions without those hypotheses to prove RH would reverse the logical direction. Here finite pairing identities and bounded-duration holomorphy are unconditional, but the infinite-tail estimate remains missing.

1.11.2. *10B: Nyman–Beurling and Báez-Duarte.* Let $\{y\} = y - \lfloor y \rfloor$. In $L^2(0,\infty)$ set

$$\rho_a(x) = \left\{ \frac{1}{ax} \right\}, \qquad \mathcal{B}_{\mathbb{N}} = \operatorname{span}\{\rho_a : a \in \mathbb{N}\}. \tag{1.50}$$

Báez-Duarte's strengthening makes RH equivalent to $\mathbf{1}_{(0,1)} \in \overline{\mathcal{B}_{\mathbb{N}}}$ [9]. The missing RH step is the approximation, not nonnegativity of a squared norm.

The present route has two actual occurrence clocks and fixed prime weights rather than freely selected approximating coefficients. Its target is nevertheless an infinite convergence statement. An arithmetic definition without mentioning zeta zeros is not unique to this construction, nor does it remove that obligation.

1.11.3. *10C: Li–Weil positivity.* Bombieri and Lagarias relate Li's criterion to general inequalities for complex multisets and to the Guinand–Weil explicit formula [8]. A positivity reformulation need not be specific to the original zeta notation. The unsupplied step is positivity for every required index or admissible test.

Ordinal pairing gives explicit finite kernels, but their automatic Gram nonnegativity is not the needed arithmetic positivity. No equality between an ordinal form and the required Li–Weil form is established here. The shift $\rho \mapsto 1 + \rho$ in (1.47) is an exact coordinate change, not a new positivity mechanism.

1.11.4. *10D: Hilbert–Pólya, Berry–Keating, and Connes.* Berry and Keating study an xp-based spectral interpretation and explicitly identify substantial speculative gaps [10]. Matching a spectral density or a semiclassical count would not identify every eigenvalue with a zeta zero. The exact divisor relation is essential.

Connes supplies a different framework: critical zeros appear as an absorption spectrum, and hypothetical off-critical zeros as resonances. His work reduces RH to a trace-formula assertion [11]. The existence of that spectral interpretation does not already exclude the resonances.

Here the starting data are discrete occurrence lists and an exact finite summation identity, without postulating an operator. This changes the starting object but supplies no self-adjoint realisation. Adopting a spectral route would still require all four obligations of Step 8C.

1.11.5. *10E: de Bruijn–Newman deformation.* In the de Bruijn–Newman family all zeros are real precisely beyond a threshold Λ. RH is equivalent to $\Lambda \leq 0$. Rodgers and Tao prove $\Lambda \geq 0$ [12]. That proves Newman's conjecture, not the complementary inequality needed for RH; it is not a defect in their theorem.

The present duration cutoff L is not a heat-flow parameter. Holomorphy for every fixed L does not give monotonic zero motion or a threshold theorem. Any such additional structure would need a proof.

1.11.6. *10F: local correlations and short intervals.* Tao's logarithmic two-point theorem and the short-interval results of Matomäki and Radziwiłłestablish genuine arithmetic cancellation in their respective regimes [5, 4]. Neither states the weighted original-order Cauchy estimate (1.39). Equation (1.28) identifies a specific unestimated height term, so describing the observable as local does not license an exchange of limits.

The distinctive interface is an exact relation between matched-pair tails and two-time heights. The inherited theorem controls its increment error, not automatically its absolute height, sign, rank weight, or full tail.

1.11.7. *10G: rigorous computation and statistical models.* Platt and Trudgian rigorously verify RH through height $3 \cdot 10^{12}$ [13]. Their theorem certifies that finite interval; it does not quantify over all heights.

Likewise, the present finite-word checks validate finite identities, and the numerical controls test mechanisms over a bounded range. Neither bounds the supremum over all $Y \geq X$ in (1.39). A statistical resemblance, or its absence, does not supply a deterministic infinite-tail estimate.

1.12. **Step 11: precise distinctiveness and limits of novelty.** The combination studied here has six demonstrated features:

(i) Ordinary primes are attached to separately numbered positive and negative occurrences of μ, with an explicit zero row.
(ii) Position, ordinal growth, and prime displacement separate exactly: the ordinal partial sum is $M(Q+1)/2$, and the excess is $D_w(N_+) - D_w(N_-)$.
(iii) Integer-clock, squarefree-clock, inverse-pair, and active-cut descriptions have exact transition identities.
(iv) Nonpolynomial duration observables distinguish paths having the same one-time height histogram.
(v) Every fixed-duration paired subseries is holomorphic for $\Re s > 1$, with an explicit cutoff-dependent majorant.
(vi) The original-order boundary term and the two-time height remainder remain visible instead of being suppressed by a rearrangement.

These features describe this construction; they do not establish priority for each ingredient. Monotone transport, layer counting, Mellin transforms, partial summation, convergence-abscissa formulae, and positivity criteria have classical antecedents. The sign flip is generic, and the bounded-duration theorem applies to much more than the Möbius word.

The complete transform is a lossless encoding, not additional independent arithmetic data: $\mu(n) = 0$ is detected by $S(n) = 1$, and otherwise $\mu(n)$ is the sign of $S(n)$. At one endpoint, $SS(n)$ depends only on $(M(n), Q(n))$. Duration data distinguish different histories, not two copies of an identical full input trajectory.

The defensible difference from the cited programmes is the particular rank-prime encoding and its exact pair/cut interfaces. No exhaustive literature theorem establishes a difference from every other construction, and no theorem here proves the remaining RH-equivalent estimate easier than the classical one. Its potential value would be a new, independently controlled arithmetic mechanism at those interfaces; that mechanism is still unresolved.

1.13. **Step 12: complete execution order and present status.**

P1. Fix clocks, weights, and finite-cut conventions: completed.
P2. Establish the initial series and $A + B + C + \delta$: completed.
P3. Separate poles of B from unproved claims about α: completed.
P4. Prove the exact pair integral and unit-cut reorganisation: completed.
P5. Prove the duration identity and order-sensitive example: completed.
P6. Determine the scope of the fixed-lag theorem: completed; an absolute-height remainder is left unestimated.
P7. Prove fixed-duration holomorphy: completed.
P8. Exclude universal tail deletion by sign antisymmetry: completed; Möbius-specific cancellation is not excluded.
P9. Establish weighted signed cancellation uniformly over durations, including unfinished pairs: unresolved.
P10. Deduce (1.39): unresolved because P9 is missing.
P11. Apply Theorem 1.10 only after proving its premise: the implication is proved; the premise is not.
P12. Alternatively prove (1.45). Contours, positivity, and spectral constructions are admissible only with their additional obligations verified; none is a completed substitute.

The next mathematical work is P9, not declaring a pole line or promoting a finite computation into an infinite bound. This chapter is an audited research programme with explicit intermediate results, not a proof of RH.

1.14. **Step 13: the energy of SS and its exact critical threshold.** For real x, the correct step-function subtraction is

$$T(x) = SS(x) - \lfloor x \rfloor + Q(x) = \mathcal{P}(N_+(x)) - \mathcal{P}(N_-(x)), \qquad \mathcal{P}(m) = \sum_{j \le m} p_j. \tag{1.51}$$

Define the nonnegative, possibly infinite quantity

$$\boxed{\mathcal{I}(\sigma) = \int_1^\infty \left(SS(x) - \lfloor x \rfloor + Q(x)\right)^2 x^{-2\sigma-1}\, dx.} \tag{1.52}$$

Using x instead of $\lfloor x \rfloor$ changes T by $-\{x\}$. Since $\{x\}$ is bounded, this change does not affect finiteness of (1.52) for $\sigma > 0$.

Proposition 1.11 (Energy comparison, including the logarithmic weight)**.** *For sufficiently large $x_0 > 1$ and any real σ,*

$$\mathcal{I}(\sigma) < \infty \quad \Longleftrightarrow \quad \int_{x_0}^\infty M(x)^2 (\log x)^2 x^{-2(\sigma-1)-1}\, dx < \infty. \tag{1.53}$$

More precisely, the two nonnegative integrands have asymptotic ratio $c_0^2 = 9/\pi^4$, wherever $M(x) \neq 0$, and both vanish when $M(x) = 0$.

Proof. The exact prime block in (1.51) has $|M(x)|$ terms. PNT gives $M(x) = o(x)$, $Q(x) \sim 2c_0 x$, and $p_j \sim j \log j$. Every index in the block is therefore $(1 + o(1))c_0 x$, uniformly over that block. Consequently

$$T(x) = c_0 x \log x\, M(x)(1 + o(1)). \tag{1.54}$$

This is a two-sided comparison in absolute value for all sufficiently large x, with the zero case treated exactly. Squaring and integrating proves the assertion. The finite initial interval causes no issue. □

Theorem 1.12 (A zero forces energy divergence even at its endpoint)**.** *If $\zeta(\rho) = 0$ for a nontrivial zero $\rho = \beta + i\gamma$, then*

$$\boxed{\mathcal{I}(\sigma) = \infty \qquad (\sigma \le 1 + \beta).} \tag{1.55}$$

In particular, unconditionally $\mathcal{I}(3/2) = \infty$.

Proof. Suppose instead that $\mathcal{I}(1 + \beta) < \infty$. By Proposition 1.11,

$$J_\beta := \int_{x_0}^{\infty} M(x)^2 (\log x)^2 x^{-2\beta-1}\, dx < \infty. \tag{1.56}$$

For $\varepsilon \ge 0$, Cauchy–Schwarz gives the uniform estimate

$$\begin{aligned} \left| \int_{x_0}^{\infty} M(x) x^{-\beta-\varepsilon-i\gamma-1}\, dx \right| &\le J_\beta^{1/2} \left(\int_{x_0}^{\infty} \frac{x^{-2\varepsilon-1}}{(\log x)^2}\, dx \right)^{1/2} \\ &\le \frac{J_\beta^{1/2}}{\sqrt{\log x_0}}. \end{aligned} \tag{1.57}$$

The same bound with absolute values inside the first integral proves absolute convergence at the boundary. The full integral $F(z) = \int_1^\infty M(x) x^{-z-1}\, dx$ is holomorphic for $\Re z > \beta$ and remains bounded as $z = \rho + \varepsilon$ tends to ρ from the right. On $\Re z > 1$, it equals $1/(z\zeta(z))$. The identity theorem, applied to $z\zeta(z)F(z) = 1$ away from $z = 1$, propagates that identity throughout $\Re z > \beta$. It either already excludes a zero in this open half-plane, or contradicts boundedness when $z \to \rho$. Thus the endpoint integral diverges. For $\sigma \le 1 + \beta$, its integrand is at least that of the endpoint because $x \ge 1$. Existence of a critical-line zero gives the final claim. □

Corollary 1.13 (RH expressed as the energy of SS)**.**

$$\boxed{\text{RH} \iff \mathcal{I}(\sigma) < \infty \text{ for every } \sigma > 3/2.} \tag{1.58}$$

More generally,

$$\inf\{\sigma \in \mathbb{R} : \mathcal{I}(\sigma) < \infty\} = 1 + \Theta. \tag{1.59}$$

Proof. Under RH, $M(x) \ll_\epsilon x^{1/2+\epsilon}$. For a given $\sigma > 3/2$, choose $\epsilon < \sigma - 3/2$ and use (1.53). Conversely, a zero with $\beta > 1/2$ contradicts the claimed finiteness at any $3/2 < \sigma \le 1 + \beta$, by Theorem 1.12. The functional equation then gives RH. The same argument with $M(x) \ll_\epsilon x^{\Theta+\epsilon}$ proves (1.59). If the supremum Θ is attained by a zero, the integral diverges at $1 + \Theta$; no endpoint assertion is inferred from a supremum that is not known to be attained. □

This proof transfers through M using an explicit two-sided comparison; it does not assume a pole correspondence for the Mellin transform of T. Indeed the exact initial-half-plane identity is

$$s \int_1^\infty T(x) x^{-s-1}\, dx = \alpha(s) - A(s) = B(s) + C(s) + \delta(s). \tag{1.60}$$

The term δ cannot be silently included in a holomorphic remainder. The logarithmic factor is also useful: it makes the boundary Cauchy–Schwarz integral $\int dx/(x \log^2 x)$ finite in (1.57). Off-boundary finiteness for every $\sigma > 3/2$ does not impose simplicity of zeta zeros; it is different from a bounded critical mean-square condition.

1.15. **Step 14: explicit poles and an attempted isolation of delta's singularities.** This step separates the meromorphic blocks from singularity claims that still depend on the remainder.

1.15.1. *14A: what the two explicit zeta blocks actually imply.* The pole statement for B is Proposition 1.2. For A, the pole at $s = 1$ has residue $1 - 1/\zeta(2)$. At $s_0 = \rho/2$, where ρ is a nontrivial zero of multiplicity m, the potential pole order is

$$\max\{m - \operatorname{ord}_{s=\rho/2} \zeta(s), 0\}. \tag{1.61}$$

For a simple zero with $\zeta(\rho/2) \neq 0$ its residue is

$$\operatorname*{Res}_{s=\rho/2} A(s) = -\frac{\zeta(\rho/2)}{2\zeta'(\rho)}. \tag{1.62}$$

At negative odd integers A has simple poles arising from $\zeta(2s)$; at negative even integers the numerator's simple zero cancels the denominator's simple zero. Both $s = 0$ and $s = 1/2$ are regular. These are poles of the displayed block, not assertions about their survival in α outside the established common domain. In particular, claiming that all poles lie on $3/2$ would already require specifying nontrivial poles and proving what happens to the other blocks.

1.15.2. *14B: a differentiated ordinal block.* The partial sum of the coefficients of δ is exactly

$$W(x) = D_w(N_+(x)) - D_w(N_-(x)) = T(x) - \frac{M(x)(Q(x)+1)}{2}. \tag{1.63}$$

For $\Re s > 2$, introduce

$$F_\zeta(z) = \frac{1}{z\zeta(z)}, \qquad P_\delta(s) = -c_0 s F'_\zeta(s-1) = \frac{B(s)}{s} - B'(s). \tag{1.64}$$

Differentiation under the absolutely convergent Mellin integral gives the exact decomposition

$$\delta(s) = P_\delta(s) + E_\delta(s), \tag{1.65}$$

$$E_\delta(s) = s\int_1^\infty (W(x) - c_0 x \log x\, M(x)) x^{-s-1}\, dx. \tag{1.66}$$

The factor $B(s)/s$ has its natural removable interpretation at $s = 0$.

Proposition 1.14 (Explicit higher-order pole candidates)**.** *At $s_0 = 1 + \rho$, if ρ is a zero of multiplicity m, the explicit meromorphic function P_δ has a pole of order $m + 1$. For a simple zero, with $r_\rho = c_0(1+\rho)/(\rho\zeta'(\rho))$,*

$$P_\delta(s) = \frac{r_\rho}{(s-s_0)^2} + \frac{r_\rho/s_0}{s-s_0} + O(1). \tag{1.67}$$

Proof. Insert $B(s) = r_\rho/(s-s_0) + b_0 + O(s-s_0)$ into $B(s)/s - B'(s)$. In higher multiplicity, differentiation raises the leading pole order by one with a nonzero leading coefficient. □

This locates poles of P_δ, and supplies a precise candidate principal part for δ. It does not establish any pole of δ. For example, holomorphy of E_δ in a neighbourhood of s_0 would make (1.67) the principal part of δ there. No such neighbourhood is presently proved. The comparison $W(x) = c_0 x \log x\, M(x)(1 + o(1))$ alone gives no power saving in the Mellin remainder. The integral (1.66) is justified for $\Re s > 1 + \Theta$ by the classical growth bound, but this does not cross the candidate singularities.

For orientation, the first five positive critical ordinates give the following numerical coefficients. These are evaluations of the explicit blocks, not a computation of a continuation of δ.

γ in $\rho = 1/2 + i\gamma$	$\Re r_\rho$	$\Im r_\rho$
14.1347251417	0.3751535752	−0.0871438181
21.0220396388	0.2639742227	0.0461618613
25.0108575801	0.2065891363	−0.0811247572
30.4248761259	0.2042801902	0.1128180994
32.9350615877	0.1811319824	−0.1250906057

At each listed $s_0 = 3/2 + i\gamma$, r_ρ is the residue of B and the double-pole coefficient of P_δ. The accompanying code evaluates the Laurent limits independently.

1.15.3. *14C: why a relative asymptotic does not prove meromorphy.* Here is an explicit control example, unrelated to any assumed formula for the true W. For $x \geq e$ take $W_{\mathrm{mod}}(x) = x^{s_0}(\log x + \log\log x)$. The second term is $o(x^{s_0}\log x)$ in modulus. Nevertheless, putting $z = s - s_0$ and $\Re z > 0$ gives

$$\begin{aligned}\int_e^\infty W_{\mathrm{mod}}(x)x^{-s-1}\,dx &= \int_1^\infty e^{-zu}(u+\log u)\,du \\ &= \frac{e^{-z}(1+z)}{z^2} + \frac{E_1(z)}{z},\end{aligned} \tag{1.68}$$

where the exponential integral satisfies $E_1(z) = -\gamma_E - \log z + O(z)$ near zero in a slit domain [16]. Thus a double-pole main term can coexist with a logarithmic branch singularity in a relatively smaller remainder. An actual meromorphic continuation for δ must exclude such possibilities by additional information, rather than by interpreting an asymptotic logarithmic weight as an exact pole order.

1.16. **Step 15: audit of five routes suggested by the Connes survey.** The source for this comparison is Connes's 2026 survey, specifically Sections 2.1.3, 2.2.3–2.2.4, 4.1, and 6.4–6.6 [15]. Its conditional programmes and proved equivalences are not failed theorems. The following distinctions determine which parts transfer to the present object.

1.16.1. *15A: stability of poles is local and conditional.* If F is meromorphic near s_0 and H is holomorphic there, the negative powers of the Laurent series of $F+H$ equal those of F. This elementary fact proves stability of every pole and its principal part under a holomorphic additive perturbation. The Borchsenius–Jessen result discussed in the survey illustrates very different behaviour for zeros after adding a nonzero constant. It does not prove that every conceivable perturbation destroys a zero alignment.

For $\alpha = B + (A + C + \delta)$, the stability argument is available precisely where $A + C + \delta$ is known to be holomorphic. It supplies no holomorphy of that perturbation and no unconditional bijection from the zeros of ζ to the poles of α. The exact pole bijection currently belongs to B.

1.16.2. *15B: Voronin universality does not forbid a polynomial lower bound.* Voronin's theorem gives approximation to fixed nonvanishing targets on suitable compact subsets of $1/2 < \Re z < 1$ by vertical translates of ζ. The approximation tolerance and the translation height are different quantifiers. Qualitative universality does not assert an error smaller than every inverse power of that height.

There is a concrete countercheck to the proposed obstruction. Euler summation gives

$$\zeta(z) = \frac{z}{z-1} - z\int_1^\infty \{x\}x^{-z-1}\,dx \qquad (\Re z > 0,\ z \neq 1). \tag{1.69}$$

On $3/2 + \eta \leq \Re s \leq 2 - \eta$, for $0 < \eta < 1/4$ and $|\Im s| \geq 1$, it implies $|\zeta(s-1)| \leq C_\eta(1 + |\Im s|)$. Hence at every regular point of B in this closed substrip,

$$\boxed{|B(s)| \geq \frac{c_\eta}{1+|\Im s|}.} \tag{1.70}$$

Thus the factor containing $1/\zeta(s-1)$ actually has a polynomial lower bound compatible with universality. This proves nothing comparable for the sum α, where cancellation between summands is possible. It does disprove the assertion that universality alone closes the entire contour route. The other contour obligations of this chapter remain.

1.16.3. *15C: a finite prime window is not yet Weil positivity.* Use the symmetric multiplicative convention of the survey's Section 4.1: $g^*(x) = \overline{g(x^{-1})}$, Haar measure dx/x, and Mellin–Fourier transform $\widehat{g}(t) = \int_0^\infty g(x)x^{-it}\,dx/x$. The Weil criterion has the form

$$\sum_v W_v(g * g^*) \le 0 \quad \text{for all admissible } g, \qquad \widehat{g}(i/2) = \widehat{g}(-i/2) = 0, \tag{1.71}$$

with the survey's sign convention. Its nonarchimedean terms are

$$W_p(f) = (\log p) \sum_{m\ge 1} p^{-m/2}(f(p^m) + f(p^{-m})). \tag{1.72}$$

If g is supported in $[\lambda^{-1}, \lambda]$, then $g * g^*$ is supported in $[\lambda^{-2}, \lambda^2]$; only prime powers $p^m \le \lambda^2$ occur. The archimedean term is still essential, and RH requires all windows and all admissible tests.

By contrast, $E_{\mathrm{mid}} = \mathrm{sgn}(M)\sum_i \tau_i g_i$ is linear in the consecutive prime gaps g_i. Its square is a rank-one quadratic expression in that gap vector. It is not the Weil form merely because only finitely many primes occur. Here τ_i denotes the triangular gap weight of the residual, distinct from $w_j = p_j - j$. Even asking whether a finite vector belongs to the domain of QW_λ requires an embedding into the function space. A quadratic form is not itself a subspace to which a vector can be orthogonal.

A well-posed finite test would specify a linear map

$$\mathcal{J}_\lambda : \mathbb{R}^k \longrightarrow \{g \in C_c^\infty((\lambda^{-1}, \lambda)) : \widehat{g}(\pm i/2) = 0\} \tag{1.73}$$

and calculate the pulled-back matrix $H_{ij} = QW_\lambda(\mathcal{J}_\lambda e_i, \mathcal{J}_\lambda e_j)$. Then $\tau^{\mathsf{T}} H \tau$ is defined and checkable. No canonical embedding, equality with E_{mid}, or implication covering all admissible tests is established by the finite-window observation. That is the precise missing identification.

1.16.4. *15D: an exact determinant with entries zero or signed primes.* The proposed finite algebraic construction is possible. First recall why the classical Redheffer determinant works. Let $Z_{ij} = \mathbf{1}_{i|j}$ for $1 \le i, j \le n$. It is upper triangular with determinant 1, and $(Z^{-1})_{ij} = \mu(j/i)$ when $i \mid j$, zero otherwise. Replacing its first column by the all-ones vector gives $R_n = Z + (\mathbf{1} - e_1)e_1^{\mathsf{T}}$. The determinant lemma yields

$$\det R_n = 1 + e_1^{\mathsf{T}} Z^{-1}(\mathbf{1} - e_1) = \sum_{j\le n} \mu(j) = M(n). \tag{1.74}$$

This is the classical divisor-incidence construction recalled in [15].

For the present transform, a different elementary encoding suffices. Put

$$U = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}, \qquad U^{-1} = \begin{pmatrix} 5 & -3 \\ -3 & 2 \end{pmatrix}, \qquad \det U = 1. \tag{1.75}$$

Given column blocks $u_i, v_i \in \mathbb{Z}^2$, form

$$\mathcal{R}(u, v) = \begin{pmatrix} U & 0 & \cdots & 0 & u_1 \\ 0 & U & \cdots & 0 & u_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & U & u_m \\ v_1^{\mathsf{T}} & v_2^{\mathsf{T}} & \cdots & v_m^{\mathsf{T}} & 0 \end{pmatrix}. \tag{1.76}$$

Theorem 1.15 (Prime-entry determinant representation)**.** *There are explicit integer matrices $\mathcal{R}_T(n)$ of size $2Q(n)+1$ and $\mathcal{R}_{SS}(n)$ of size $2n+1$, all of whose entries belong to $\{0\}\cup\{\pm p_j : j \ge 1\}$, such that*

$$\boxed{\det \mathcal{R}_T(n) = T(n), \qquad \det \mathcal{R}_{SS}(n) = SS(n).} \tag{1.77}$$

Proof. The Schur complement of the diagonal block $D = \operatorname{diag}(U, \dots, U)$ gives

$$\det \mathcal{R}(u, v) = -\sum_{i=1}^{m} v_i^{\mathsf{T}} U^{-1} u_i. \tag{1.78}$$

For any zero or signed prime t_i, choose $u_i = (t_i, 0)^{\mathsf{T}}$ and $v_i = (-2, -3)^{\mathsf{T}}$. Then $v_i^{\mathsf{T}} U^{-1} u_i = (-10 + 9)t_i = -t_i$. For $\mathcal{R}_T(n)$ use the $Q(n)$ signed-prime summands $S(s_q)$.

For $\mathcal{R}_{SS}(n)$ use one block per integer $i \le n$. If $\mu(i) \neq 0$, use the preceding choice with $t_i = S(i)$. If $\mu(i) = 0$, choose instead $u_i = (5, 7)^{\mathsf{T}}$ and $v_i = (-2, -7)^{\mathsf{T}}$. In this case $U^{-1} u_i = (4, -1)^{\mathsf{T}}$ and $-v_i^{\mathsf{T}} U^{-1} u_i = 1 = S(i)$. Every entry is zero or a signed prime, and summing the contributions in (1.78) proves both identities. □

For example, at $n = 4$ the transformed sequence is $(2, -2, -3, 1)$, giving $\det \mathcal{R}_T(4) = -3$ and $\det \mathcal{R}_{SS}(4) = -2$.

This settles the existence question, but the block construction encodes any sum of signed primes, with a separate gadget for a unit summand. It is not a newly proved divisor-incidence constraint. The already established transfer gives

$$\text{RH} \iff |\det \mathcal{R}_T(n)| \ll_\epsilon n^{3/2+\epsilon} \quad \text{for every } \epsilon > 0. \tag{1.79}$$

The determinant growth estimate remains unproved. Neither the representation nor the general Hadamard determinant inequality supplies this bound. The construction is finite and testable; its RH-strength estimate still requires the original arithmetic cancellation.

1.16.5. *15E: what regularising the pair measure actually gives.* The Guinand–Weil formula applies to an admissible test function. A signed atomic measure is not such a function until regularised. The finite regularisation can be worked out exactly, with no continuation of δ assumed. Set $c_n = \mu(n) w_{j(n)}$ at squarefree n and $c_n = 0$ otherwise, and

$$\nu_X = \sum_{n \le X} c_n \delta_{\log n}, \qquad h_{X,\varepsilon} = \varphi_\varepsilon * \nu_X, \qquad f_{X,\varepsilon,s}(x) = x^{-s} h_{X,\varepsilon}(\log x), \tag{1.80}$$

where $\varphi_\varepsilon \in C_c^\infty(\mathbb{R})$. This ν_X retains unfinished pairs; it is the logarithmic pushforward of the original-order coefficient measure. Write

$$\Phi_\varepsilon(v) = \int_{\mathbb{R}} \varphi_\varepsilon(u) e^{vu}\, du, \qquad \delta_X(s) = \sum_{n \le X} c_n n^{-s}. \tag{1.81}$$

For the unnormalised Mellin convention $\widetilde{f}(z) = \int_0^\infty f(x) x^{z-1}\, dx$, direct substitution gives

$$\widetilde{f}_{X,\varepsilon,s}(z) = \Phi_\varepsilon(z - s)\delta_X(s - z). \tag{1.82}$$

Using the survey's Section 2.1.3 convention, the explicit formula becomes

$$\sum_\rho \Phi_\varepsilon(\rho - s)\delta_X(s - \rho) = \Phi_\varepsilon(-s)\delta_X(s) + \Phi_\varepsilon(1 - s)\delta_X(s - 1) - \sum_v W_v(f_{X,\varepsilon,s}). \tag{1.83}$$

Zeros are counted with multiplicity. At fixed X, ε, s the smooth compact support gives rapid vertical decay of the Mellin transform, so the zero sum is legitimate. Here the unnormalised local terms are

$$W_p(f) = (\log p) \sum_{m \ge 1} \left(f(p^m) + p^{-m} f(p^{-m})\right); \tag{1.84}$$

the $p^{-m/2}$ in (1.72) belongs to the different, symmetric normalisation.

Equation (1.83) is a proved finite regularised identity. It is not a closed continuation formula for $\delta(s)$: it also contains $\delta_X(s - 1)$, all the shifted values $\delta_X(s - \rho)$, and local terms depending on the regularisation and cutoff. Passing to $X \to \infty$ and then removing the regularisation requires uniform

summability or another justified limiting procedure. That arithmetic estimate is not supplied by admissibility of each finite test. This route remains meaningful, with the limiting step explicit, rather than bypassing the unresolved control of δ.

1.16.6. *15F: the survey's current programme and the remaining work here.* The survey's Sections 6.4–6.6 distinguish several established constructions from further obligations: a simplicity and evenness property for the lowest eigenfunction of the truncated Weil form, and sufficient control relating that eigenfunction to another family of approximants. Convergence for one family cannot simply be transferred to another because they appear related. This comparison is recorded without pursuing an operator route in the present programme.

The same logical issue occurs here: every fixed-duration $\delta_{\leq L}$ can be holomorphic in a large half-plane while the limit as $L \to \infty$ has singularities. The new determinant identity, the finite Guinand–Weil identity, and the energy equivalence are established intermediate statements. None supplies the missing uniform arithmetic estimate.

What remains unresolved is the arithmetic estimation of the signed, weighted two-time terms, including unfinished pairs. The two-cut identity is exact; by itself it does not control δ. The explicit poles of B and P_δ have been located. The singularities of the full δ, and their effect on α, remain unproved without control of E_δ or another valid continuation mechanism.

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